Anchoring Bias in the NFL Gambling Market

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Abstract

We explore the impact of the anchoring bias in the NFL betting market to overcome limitations within financial literature regarding investor behavior. Consistent with an anchoring bias, we find bettors prefer to bet on teams with better pre-season Super Bowl odds throughout the entire season. We further document that this strategy has a significant inverse relationship to profitability in week 1 of the NFL season. Furthermore, we show sportsbooks continue to incorporate pre-season odds into their closing lines throughout the entire season, suggesting they may anchor to pre-season odds as well. These results have important implications for financial markets.

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1. Introduction

The financial literature documents that investors exhibit a cognitive bias known as anchoring bias, whereby they disproportionately rely on initial information and fail to adjust adequately when new information arises. This behavior, stemming from limitations in cognitive resources, contributes to market inefficiencies and is a critical consideration within psychology-based asset pricing theory (Hirshleifer, 2001). Research shows the anchoring bias hinders price adjustments (Birru, 2015) resulting in mispricing (Li et al., 2023). In this paper, we explore the incidence and consequences of the anchoring bias in the NFL gambling setting to overcome common limitations in finance literature such as the joint hypothesis issue and unknown investment time horizons while sharing features with the stock market like arbitrage activity, information availability, and transaction volume (Moskowitz, 2021). Specifically, we examine the impact of pre-season Super Bowl odds on gambler betting preferences and profitability throughout the entire season. We rely on these pre-season odds as they are a proxy for season long expectations for each team.

2. Data

Data containing closing line, final score, and betting percentages is collected from SportsInsights from 2003 to 2023 following previous literature (e.g., Shank, 2022 or Sung & Tainsky, 2014). Pre-season Super Bowl odds are collected from Pro-Football-Reference.com.[1] On average, 47.9% of wagers are on the home team while the home team covers the spread 48.7% of the time. Super Bowl odds vary between +250 to +100,000. We calculate the difference between the home and away odds to determine an anchoring baseline. We also examine the percentage of games the previous week that resulted in the home team covering the spread (48.5%) following

[1] Other papers to employ this source include Pitts and Evans (2020), Roach (2023), Paul et al. (2023), and others.

Nofsinger and Shank (2023) and the difference between how home and away teams performed against the spread the previous week (0.0694) following Durand et al. (2021) as research documents these as important factors in determining wager preferences related to the recency bias. We also include the season to date difference between home and away teams in the number of outright wins and the difference in games they covered the spread to control for realized team performance throughout the season.[2]

Table 1: Summary Statistics
This table presents the summary statistics for the sample, N=5,088.

Variables	Mean	SD	Min	Max
Home Bet Percentage	0.479	0.162	0.0800	0.840
Home Cover	0.487	0.500	0	1
Home - Away Super Bowl Odds	6.170	10,099	-99,725	99,725
Weekly Aggregate Home Cover $_{t-1}$	0.485	0.126	0.167	0.867
Home Points Spread Covered - Visitor Points Spread Covered t-1	0.0694	18.66	-68	63.50
Home - Away Outright Wins	-0.0573	2.644	-11	13
Home - Away Spread Cover Wins	0.0140	1.851	-10	8

3. Results

Figure 1 provides a graphical illustration of how pre-season Super Bowl odds influence wagering decisions based on 5 quintiles.[3] Consistent with an anchoring bias, gamblers prefer to bet on the home (away) team if the home team are expected to do better (worse) for the season. This preference is most substantial in week 1 and declines in following weeks, but still deviates from 50% for the full season, excluding quintile 4.

Insert Figure 1 here

[2] To our knowledge no other paper has incorporated this into their models.
[3] Quintiles based on the Home – Away Super Bowl Odds variable can be categorized as Q1 (<-3500, n ≈ 23%), Q2 (>=-3500 & <= -500, n ≈ 20%), Q3 (>-500 & < 500, n ≈14%), Q4(>= 500 & <=3500, n ≈ 20%), and Q5(>3500 ≈ 23%)

Table 2 shows how pre-season Super Bowl odds impact gambler wagering decisions.[4] Column 1 shows that the difference between pre-season odds has a significant influence on wagering preference. Specifically, if the home team is expected to perform worse in the season compared to the visiting team, fewer wagers are placed on the home team. Columns 3 and 5 show this anchoring bias is consistent at the 1% level through weeks 2 and 8 and through weeks 9 to 17 separately. While a recency bias may be able to explain why bettors wager on the better team at the beginning of the season, our results support the hypothesis that gamblers anchor to pre-season expectations throughout the first half of the season after controlling for their in-season winning percentage.

Table 2: Betting Percentages Based on Pre-Season Super Bowl Odds

This table presents OLS regression results where the dependent variable is the percentage of wagers placed on the home team. T-statistics are presented under the coefficients with significance displayed at the 10% (*), 5% (**), and 1% (***) levels.

VARIABLES	Week 1	Week 1	Weeks 2-8	Weeks 2-8	Weeks 9-17	Weeks 9-17
Home - Away Super Bowl Odds	-0.0009***		-0.0004***		-0.0001***	
	(-10.217)		(-14.696)		(-2.881)	
Home - Away Super Bowl Odds Quintile 1		7.1162***		8.5139***		2.2547***
		(2.972)		(9.432)		(2.981)
Home - Away Super Bowl Odds Quintile 2		2.0115		6.0065***		3.1634***
		(0.817)		(6.406)		(4.205)
Home - Away Super Bowl Odds Quintile 4		-16.5667***		-5.6644***		-3.4622***
		(-6.752)		(-6.074)		(-4.554)
Home - Away Super Bowl Odds Quintile 5		-23.0970***		-10.7343***		-3.3426***
		(-9.930)		(-11.657)		(-4.442)
Home Points Spread Covered - Visitor Points Spread Covered $_{t-1}$			0.0776***	0.0873***	0.1157***	0.1158***
			(4.549)	(5.606)	(9.440)	(9.651)
Weekly Aggregate Home Cover $_{t-1}$			2.9756	5.9694***	2.6458	2.8208
			(1.305)	(2.857)	(1.496)	(1.628)

[4] We do not include a home underdog or closing spread variable as in Nofsinger and Shank (2023) due to multicollinearity issues with the Super Bowl odds variable

Home - Away Outright Wins			3.7938***	2.3368***	2.7197***	2.3431***
			(14.489)	(9.369)	(29.027)	(24.228)
Home - Away Spread Cover Wins			0.9786***	1.9365***	0.6661***	0.9958***
			(3.236)	(6.923)	(4.987)	(7.474)
Observations	319	319	2,123	2,123	2,935	2,935
R-squared	0.248	0.496	0.331	0.443	0.454	0.477

Given the large variability of the Home – Away Super Bowl Odds variable as shown by the standard deviation in Table 1, we employ a quintile approach to control for outliers. The results show gamblers are more averse to betting on the home team when they are expected to have a lesser chance of winning the Super Bowl (i.e., Quintile 4 and 5) while preferring to bet on the home team when they are favored to have a better season (i.e., Quintile 1 and 2) and is consistent throughout the entire season. These results are significant at the 1% level in nearly all models. When examining our control variables, we support previous research showing that performance against the spread in the previous week greatly impact wagering decisions and provide new evidence that straight up wins and season to date spread covering plays a large role in wagering decisions. Using the quintile approach along with controls, our model can explain over 40% of wagering decisions in week 1, weeks 2-8 and weeks 9-17.

Table 3 examines market efficiency given the observed extreme wagering preferences by employing a Probit regression where the dependent variable is a binary variable that equals 1 if the home team covers the spread and 0 otherwise. Levitt (2004) argues sportsbooks use their expertise to maximize their profits. As such, we would expect to see signs opposite betting preferences and profitability. Column 2 shows betting against the home team when they are expected to do better than the visiting team throughout the season is negatively, and statistically significantly, related to the home team covering the spread in the first week of the season. As such, it appears sportsbooks may use gamblers' biased wagers against them. Furthermore, we find no significance for other weeks throughout the season for our variables of interest. We do, however, show that in the second half of the season, teams that covered the spread more often previously were significantly less likely to cover the spread. We believe this result is new to the literature and may warrant further research.

Table 3: Betting Outcomes Based on Pre-Season Super Bowl Odds
This table presents OLS regression results where the dependent variable is a dichotomous variable that equals 1 if the home team covered the spread. T-statistics are presented under the coefficients with significance displayed at the 10% (*), 5% (**), and 1% (***) levels.

VARIABLES	Week 1	Week 1	Weeks 2-8	Weeks 2-8	Weeks 9-17	Weeks 9-17
Home - Away Super Bowl Odds	0.0007		-0.0001		0.0002	
	(0.946)		(-0.496)		(0.658)	
Home - Away Super Bowl Odds Quintile 1		-41.9*		7.5605		-8.9164
		(-1.765)		(0.805)		(-1.076)
Home - Away Super Bowl Odds Quintile 2		-29.1		0.8176		-9.8511
		(-1.190)		(0.084)		(-1.196)
Home - Away Super Bowl Odds Quintile 4		-24.6		-0.4332		-10.5651
		(-1.020)		(-0.045)		(-1.272)
Home - Away Super Bowl Odds Quintile 5		-23.3		-3.0204		-5.5239
		(-1.014)		(-0.315)		(-0.671)
Home Points Spread Covered - Visitor Points Spread Covered $_{t-1}$			-0.2419	-0.2364	-0.0198	-0.0217
			(-1.502)	(-1.467)	(-0.151)	(-0.165)
Weekly Aggregate Home Cover $_{t-1}$			31.7991	32.7595	6.3578	5.3944
			(1.467)	(1.504)	(0.336)	(0.285)
Home - Away Outright Wins			0.8128	-0.0309	0.7622	0.6849
			(0.328)	(-0.012)	(0.763)	(0.650)
Home - Away Spread Cover Wins			-4.6425	-4.0849	-2.6018*	-2.5195*
			(-1.611)	(-1.399)	(-1.820)	(-1.731)
Observations	327	327	2,102	2,102	2,846	2,846

Next, we examine the impact of pre-season odds on closing lines produced by sportsbooks to examine if anchoring to pre-season odds is present. In Table 4 we run the same model as previous tables where the dependent variable is the closing line where -10 would be considered a greater favorite compared to -2 from the home team's perspective. Columns 1 and 2 show that sportsbooks are more likely to make the home team a greater favorite if they are expected to have a better season, which is intuitive given it is the first week of the season. However, remaining results show sportsbooks continue to incorporate pre-season odds in closing lines throughout the entire season even after accounting for recent performance, and season to date differences in wins and spread coverage. One explanation is sportsbooks are following bettors' wagers when setting and adjusting their lines. However, Levitt (2004) argues that since sportsbooks are more informed, they should use gamblers biases against them. As such, the results may suggest sportsbooks continue to anchor to pre-season odds throughout the entire season with respect to setting and adjusting spreads.

Table 4: Closing Spread Based on Pre-Season Super Bowl Odds
This table presents OLS regression results where the dependent variable is the closing spread from the home team's perspective. T-statistics are presented under the coefficients with significance displayed at the 10% (*), 5% (**), and 1% (***) levels.

VARIABLES	Week 1	Week 1	Weeks 2-8	Weeks 2-8	Weeks 9-17	Weeks 9-17
Home - Away Super Bowl Odds	0.0003***		0.0002***		0.0001***	
	(15.740)		(28.728)		(16.034)	
Home - Away Super Bowl Odds Quintile 1		-4.17***		-4.25***		-2.28***
		(-8.402)		(-17.468)		(-9.688)
Home - Away Super Bowl Odds Quintile 2		-2.00***		-1.91***		-0.99***
		(-3.941)		(-7.580)		(-4.202)
Home - Away Super Bowl Odds Quintile 4		3.65***		1.90***		1.24***
		(7.219)		(7.568)		(5.226)
Home - Away Super Bowl Odds Quintile 5		6.59***		4.61***		2.71***
		(13.638)		(18.601)		(11.554)
Home Points Spread Covered - Visitor Points Spread Covered $_{t-1}$			-0.0152***	-0.02***	-0.0311***	-0.03***
			(-3.182)	(-4.666)	(-7.953)	(-8.256)
Weekly Aggregate Home Cover $_{t-1}$			-0.7227	-1.82***	0.3428	0.14

			(-1.119)	(-3.211)	(0.608)	(0.261)
Home - Away Outright Wins			-1.9543***	-1.45***	-1.2805***	-1.12***
			(-26.325)	(-21.483)	(-42.889)	(-37.271)
Home - Away Spread Cover Wins			0.0989	-0.21***	-0.1331***	-0.27***
			(1.153)	(-2.806)	(-3.128)	(-6.423)
Observations	335	335	2,153	2,153	2,935	2,935
R-squared	0.427	0.703	0.567	0.670	0.646	0.675

4. Conclusion

This paper is the first to examine the impact of the anchoring bias in sports betting. We document that bettors prefer to wager on teams that are expected to do better at the beginning of the season. This preference is strongest at the beginning of seasons but persists throughout the entire season. Sportsbooks take advantage of this bias as it is negatively related to winning in the first week of the season but we do not find a relationship with profitability over the remainder of the season. Furthermore, we document that pre-season Super Bowl odds have a significant relationship to the final line set by the sportsbooks, suggesting they may be susceptible to the anchoring bias as well.

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Figure 1: Betting Percentages by Week and Quintile

This figure presents the average wager on the home team based on the quintile of the Home - Away Super Bowl Odds variable.